HDOC Day-75

Problem Statement

Q1. Create enum A, B, C with values 10, 11, 12 and print them. Q2. Use enum ADD, SUBTRACT, MULTIPLY and switch to calculate results for two integers.

Algorithm

Q1: Declare enum with A=10; print A, B and C. Q2: Read operation name; convert to enum; read two integers; switch on enum and display result. Reject invalid operations.

Test Case Table

TC	Question	Input	Expected result
1	Q1	Nil	A=10, B=11, C=12
2	Q2	ADD 10 20	30
3	Q2	SUBTRACT 20 10	10
4	Q2	MULTIPLY 5 6	30
5	Q2	ADD 0 0	0
6	Q2	SUBTRACT 5 10	-5
7	Q2	MULTIPLY 0 8	0
8	Q2	ADD -5 10	5
9	Q2	DIVIDE	Invalid operation
10	Q2	MULTIPLY -3 4	-12

C Program - Question 1

```
#include <stdio.h>
enum Values { A=10, B, C };
int main(void) {
    printf("A=%d\nB=%d\nC=%d\n", A, B, C);
    return 0;
}
```

C Program - Question 2

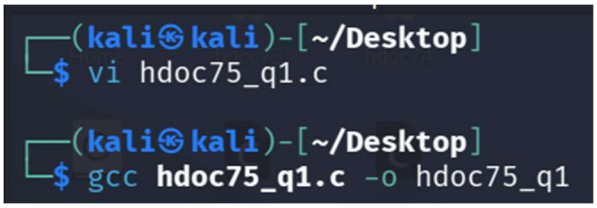
```
#include <stdio.h>
#include <string.h>
enum Operation { ADD=1, SUBTRACT, MULTIPLY };
int main(void) {
    char input[20];
    int a, b;
    enum Operation choice;
    printf("Enter operation (ADD/SUBTRACT/MULTIPLY): ");
    if (scanf("%19s", input)!=1) return 1;
```

```
    if (strcmp(input,"ADD")==0) choice=ADD;
    else if (strcmp(input,"SUBTRACT")==0) choice=SUBTRACT;
    else if (strcmp(input,"MULTIPLY")==0) choice=MULTIPLY;
    else { printf("Invalid operation.\n"); return 1; }
    printf("Enter two numbers: ");
    if (scanf("%d %d", &a, &b)!=2) return 1;
    switch(choice) {
        case ADD: printf("Result: %d\n",a+b); break;
        case SUBTRACT: printf("Result: %d\n",a-b); break;
        case MULTIPLY: printf("Result: %d\n",a*b); break;
    }
    return 0;
}
```

Compilation and Execution Commands

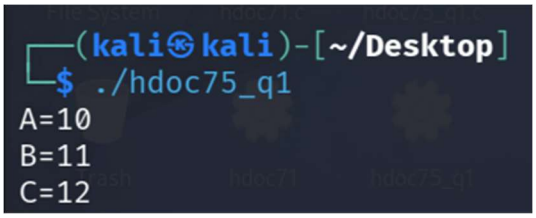
gcc hdoc75_q1.c -o hdoc75_q1
./hdoc75_q1
gcc hdoc75_q2.c -o hdoc75_q2
./hdoc75_q2

Compilation and Execution - Screenshot

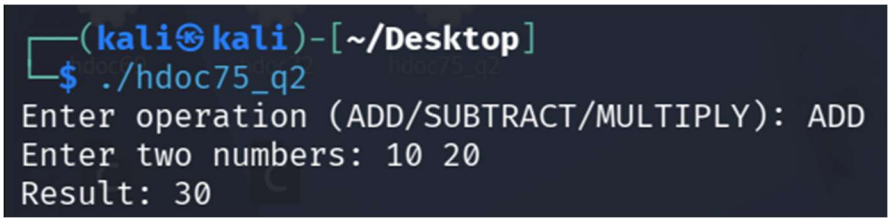




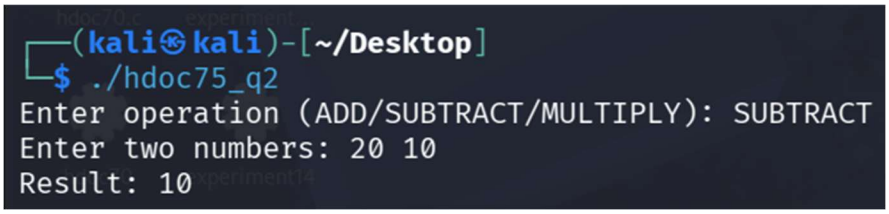
Test Case 1 - Screenshot



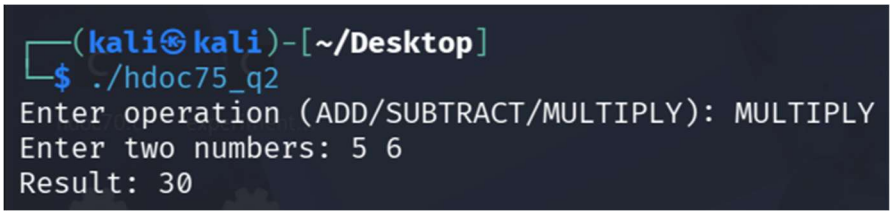
Test Case 2 - Screenshot



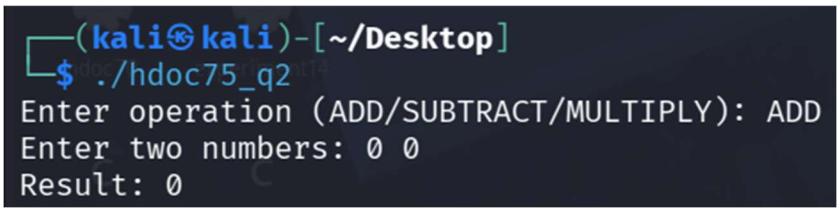
Test Case 3 - Screenshot



Test Case 4 - Screenshot



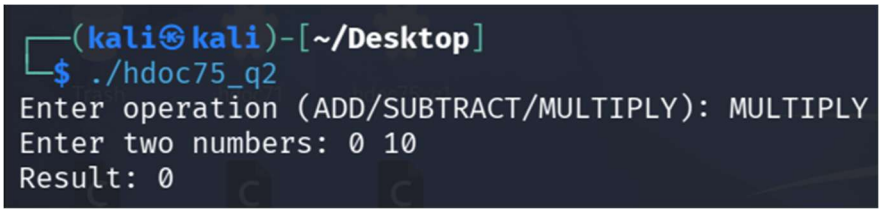
Test Case 5 - Screenshot



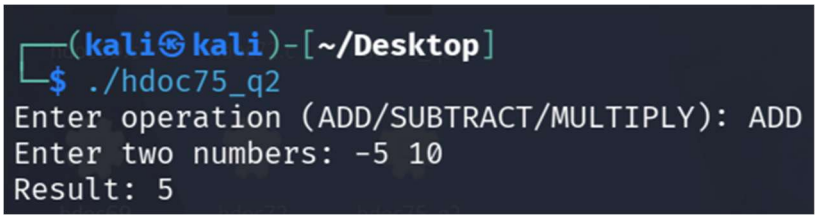
Test Case 6 - Screenshot



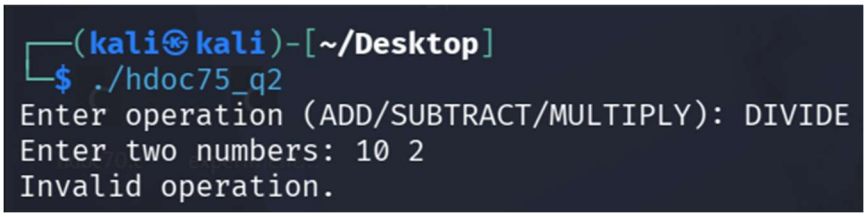
Test Case 7 - Screenshot



Test Case 8 - Screenshot



Test Case 9 - Screenshot



Test Case 10 - Screenshot

